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The Ranking Transfers, the Threshold Does Not: Early At-Risk Prediction in Unseen Courses

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Abstract

Early-warning models in online higher education are usually tested on splits that keep the same courses in training and test data. We ask what survives when a model is moved to a course it has never seen. Using the Open University Learning Analytics Dataset (32,593 enrolments, 7 modules), we predict withdrawal or failure from the first quarter of each course, holding out one module at a time with no student on both sides. Three common evaluation choices more than double the apparent transfer gap. Once they are corrected, the ranking of students largely transfers (average AUC loss of 0.014), but calibration does not: in one module, the transferred model flags 95% of students for an observed risk of 37%, at every threshold from 0.3 to 0.7. Signed SHAP attributions show that the model uses the same features as local models but reads engagement and early scores on absolute scales that depend on course design. Normalizing these features within the cohort, without labels, brings the alert volume close to the observed risk, as recalibration on labelled students does, with higher precision. Pooling courses also creates a gender disparity that within-course audits reveal as a composition effect.

Notes for Practice

- Early-warning models are often built on some courses and then used in others, yet they are usually evaluated on data in which the same courses appear in training and testing.
- In a course it has never seen, a model still ranks students well, but its probabilities and its number of alerts can be badly wrong: in one of the seven courses studied, it flagged 95% of the students for an observed risk of 37%.
- The cause is course design. A given number of clicks, or the absence of an early grade, does not mean the same thing in every course.
- Expressing engagement and early-assessment indicators relative to the student's cohort, which requires no outcome data, brings the number of alerts back in line with the actual risk.
- Institutions should check alert rates against expected risk, and audit fairness within each course as well as overall: pooling courses can create disparities that exist in none of them.

Keywords

Learning analytics, at-risk prediction, early warning systems, dropout, transferability, calibration, dataset shift, SHAP, fairness, OULAD

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1. Introduction

Dropout and failure remain persistent problems in online higher education. Because learning management systems (LMS) record what students do from the first day of a course, they make it possible to identify students who may need support while there is still time to help them, and early-warning systems built on these records have become common. The Open University Learning Analytics Dataset (OULAD) (Kuzilek et al., 2017) holds a central place in this research. It is open and large, it combines demographics, registration records, virtual learning environment (VLE) clickstreams and assessment results for 22 module presentations, and many predictive studies have used it (Jin et al., 2024).

By now, a high accuracy on OULAD tells us little by itself. Many studies compute predictors over the whole presentation, so that the resulting models are retrospective rather than early (Elsayed et al., 2026). And models are nearly always evaluated on splits that keep the same courses on both sides; even a calibrated early-warning study

holds out presentations rather than modules, so that other presentations of a test module may appear in training (Wu et al., 2026). Beyond OULAD, several studies have found that predictive models are largely course-specific (Conijn et al., 2017; Gašević et al., 2016; López-Zambrano et al., 2020), but few have looked at why the models fail when they are moved.

The question matters to any institution that wants to use a model in a course it was not trained on, such as a new programme with no labelled history. OULAD lets us study it methodologically, since each of its seven modules can in turn play the part of the unseen course. We ask which aspects of an early at-risk model still hold in a course it has never seen, why the others break down, and what can be done before any outcome labels are available. The paper makes five contributions.

1. *An evaluation protocol and an audit.* Three common departures from a strict protocol together more than double the apparent transfer gap.
2. *A diagnosis.* We separate module difficulty, the loss in ranking and the drift of alert volume and calibration. Ranking largely transfers; alert volume and calibration do not, whatever the threshold, learner or weighting scheme.
3. *An explanation.* Signed SHAP attributions show that the transferred model uses the same features as local models but reads them on absolute scales whose meaning depends on course design.
4. *A consequence for fairness.* When courses are pooled, calibration drift turns into a gender disparity through course composition, whereas disparities linked to deprivation are present within courses.
5. *A remedy derived from the diagnosis.* Normalizing engagement and early-assessment features within the cohort corrects the drift without labels. We compare it, on the same students, with recalibration, prevalence correction and flagging a fixed share, and we describe the conditions it requires.

Our claims differ in strength. The experiments establish the results for the students of seven modules at one university. They support, without proving it, the interpretation that cohort normalization makes a model less sensitive to course design. Whether this holds in other institutions is a hypothesis that these data cannot test.

2. Related Work

2.1 Early prediction on OULAD

OULAD was released to support the Open University's work on identifying students at risk (Kuzilek et al., 2017). Later studies have varied the observation window, predicting at several points of the course (Adnan et al., 2021) or from the first 25% of each presentation with deep architectures (Malkani et al., 2026). Hlosta et al. (2017) address a related cold-start problem by training on the current presentation instead of past data. Closest to ours, Wu et al. (2026) use a 28-day window, calibrate a gradient-boosting model, audit subgroup fairness and examine alert capacity. We hold out whole modules rather than presentations, explain why calibration fails, and evaluate a correction that uses the target cohort but no labels.

2.2 Portability of predictive models across courses

The Open Academic Analytics Initiative found that a model trained at one institution could be used at others with a moderate loss of accuracy (Jayaprakash et al., 2014). Studies at the course level are less encouraging. The predictive value of LMS indicators depends on instructional conditions (Gašević et al., 2016), models differ substantially between courses of the same institution (Conijn et al., 2017), and models moved directly from one Moodle course to another work only under some conditions (López-Zambrano et al., 2020). Others have built course-agnostic early predictors from large volumes of LMS logs (Riestra-González et al., 2021) or applied transfer learning across MOOC offerings (Boyer & Veeramachaneni, 2015). These studies ask whether accuracy carries over; we ask which part of the model fails to carry over, and why.

2.3 Dataset shift, calibration and normalization

Moving a model to a new context is a form of dataset shift, in which changes in the covariates, in the class priors or in the relation between them can all affect a classifier (Moreno-Torres et al., 2012). Clinical prediction offers a well-documented parallel: discrimination often holds up in a new setting while calibration does not, and re-estimating the intercept or the slope on local data recovers much of the loss (Janssen et al., 2008; Van Calster et al., 2019). Platt scaling (Platt, 1999) plays the same role in machine learning. If only the prevalence changes, it can be estimated

without labels, either by expectation–maximization (Saerens et al., 2002) or from the confusion matrix of a black-box classifier (Lipton et al., 2018). If the distribution of the inputs changes, domain adaptation methods re-estimate normalization statistics on the target domain (Li et al., 2018) or adapt the model at test time using unlabelled target data (Wang et al., 2021). Normalizing features within a cohort is a simple, model-agnostic version of this idea; we use it to test the mechanism we identify, not as a new technique.

2.4 Fairness of predictive student models

Fairness audits of student models usually compare error rates and alert rates between demographic groups (Baker & Hawn, 2022; Gardner et al., 2019). On OULAD, recent work reports trade-offs between accuracy and fairness (Opoku et al., 2025), regional differences in recall (Elsayed et al., 2026) and gaps in alert rates that depend on the alerting policy (Wu et al., 2026). When groups have different base rates, error-rate parity and calibration cannot all be satisfied at once (Chouldechova, 2017; Kleinberg et al., 2017), and equal opportunity (Hardt et al., 2016) focuses on true-positive rates instead. We add a structural observation: pooling several contexts can produce group disparities that exist in none of them, a case of Simpson's paradox (Simpson, 1951).

3. Data and Methods

3.1 Dataset and target

OULAD (Kuzilek et al., 2017) (CC-BY 4.0) contains 32,593 enrolments in 22 presentations of 7 modules, labelled AAA to GGG, at the Open University in the United Kingdom. Each enrolment ends with one of four outcomes: Distinction, Pass, Fail or Withdrawn. We predict the unfavourable outcome $y = 1$, defined as Withdrawn or Fail (52.8% of enrolments), because both call for the same kind of early support; the supplementary material repeats the analyses with Withdrawn alone. The study is a secondary analysis of these public, anonymized data. No new data were collected and no attempt was made to re-identify students, so no ethical approval was required.

3.2 Early window, population and features

For a presentation p lasting L_p days, the cut-off day is $d_c = \mathrm{round}(c\,L_p)$, with $c \in \{0.10, 0.20, 0.25, 0.33, 0.50\}$ and $c = 0.25$ as the reference. The population at d_c includes the students who registered on or before d_c and had not unregistered by then. Students who have already left cannot be helped by an early warning, and keeping them in the data makes prediction artificially easy (Section 4.1).

Eighteen predictors are computed for each enrolment, using only data dated on or before d_c: *demographics* (gender, region, highest education, IMD band, age band, previous attempts, studied credits, disability); *registration* (registration date relative to the course start, and days registered in advance); *VLE engagement* (total clicks, active days, distinct resources, first and last active day, clicks per active day); *early assessments* (number of assessments, and mean score of those, whose deadline *and* submission both fall on or before d_c). Missing engagement is coded as zero activity, missing first and last active days by sentinel values outside the observed range, and a missing early score as -1.

Besides excluding the final result and the unregistration date from the predictors (Kaufman et al., 2012), we test the whole pipeline for invariance: if every raw record dated after d_c (clicks, submissions, deadlines, unregistrations) is deleted, the feature table must remain exactly the same. The test detects, for example, a filter applied to assessment deadlines alone, which lets late submissions through (4.0% of the retained submissions at $c = 0.25$).

3.3 Absolute and cohort-normalized features

The *absolute* feature set uses these values directly, as most studies do. The *relative* set replaces the eight engagement and early-assessment features with their percentile rank within the presentation,

$$\tilde{x}_{ij} = \frac{1}{n_p}\Big(L_{ij} + \frac{E_{ij}+1}{2}\Big), \tag{1}$$

where P_p is the population of presentation p at d_c, $n_p = |P_p|$, L_{ij} is the number of students of P_p whose value of feature j is lower than x_{ij}, and E_{ij} the number, i included, whose value is equal to it. Tied values receive their average rank, so students without an early score share the lowest ranks, at a level that grows with their share of the cohort. Demographic and registration features are unchanged.

No labels are used, but the transformation is *transductive*. A prediction requires the cohort observed on the same date, the ranks change as students enrol or leave, and a cohort that disengages as a whole keeps the same ranks. To check whether ranks in particular are needed, we compare them with a z-score computed after winsorizing at the 1st and 99th percentiles of the cohort, and with a robust standardization based on the median and the interquartile range.

To find out how large a cohort ranks need, we recompute them in the target module within random sub-cohorts of $n \in \{10, 20, 50, 100, 200\}$ students (20 random partitions), leaving the trained model unchanged.

3.4 Models and protocols

Our main model is a random forest (Breiman, 2001) with 300 trees and balanced class weights (scikit-learn; Pedregosa et al., 2011), which flags a student when $\hat{p} \geq 0.5$. With balanced weights, this rule approximately compares the class-conditional likelihoods of the student's features, which gives an operating point that does not depend on the prevalence in the training data. Since this choice could either create or hide a calibration problem, we also report a logistic regression with balanced weights and an unweighted random forest calibrated by isotonic regression (internal three-fold cross-validation).

In protocol (A), the data are split into five folds, stratified on y and grouped by student, and the out-of-fold predictions cover every enrolment. In protocol (B), the model is trained on six modules and predicts the seventh. Because 6.8% of students are enrolled in more than one module (up to 48.7% of the students of CCC), we use a *strict* version of (B): students enrolled in the held-out module are removed from the training modules, so that, as in (A), no student appears on both sides. The training sets of the two protocols have similar sizes (the (B) set is 0.93 to 1.22 times the size of the (A) set).

In both protocols, metrics are computed on the students of each module and then averaged across modules. An AUC pooled over modules in (A) would partly reward the model for separating modules with different base rates, something (B) cannot do. For each module m, we define on the same students

$$\text{difficulty:} \quad D_m = \mathrm{AUC}_A(m), \tag{2}$$

$$\text{ranking cost:} \quad C_m = \mathrm{AUC}_A(m) - \mathrm{AUC}_B(m), \tag{3}$$

$$\text{flag-rate drift:} \quad \Delta_m^{\text{same}} = \bar{f}_B(m) - \bar{f}_A(m), \tag{4}$$

where $\bar{f}(m)$ is the share of the module's students who are flagged, both terms coming from the same model and feature set. To compare feature sets or remedies with one another, we also use a common reference, $\Delta_m^{\text{ref}} = \bar{f}(m) - \bar{f}_A^{\text{abs}}(m)$, the difference from the absolute-feature model that has seen the module. That reference itself flags too many students in some modules, so our main criterion for alert volume is the distance $|\bar{f}(m) - \bar{y}(m)|$ between the share flagged and the observed at-risk rate.

3.5 Calibration, remedies and uncertainty

For each module and both protocols, we report the Brier score, the mean predicted risk, calibration-in-the-large (the intercept of a logistic regression of y with $\mathrm{logit}(\hat{p})$ as an offset), the calibration slope and reliability diagrams. Because of the balanced weights, the probabilities are not calibrated even in (A). What we measure is therefore the *degradation* of calibration under transfer, the difference (B) − (A) on the same students. We also compute the drift for thresholds from 0.3 to 0.7.

We compare several ways of flagging the students of a held-out module: the raw transferred model; *Platt recalibration* (Platt, 1999) of that model on labelled students from the target module, and its cheaper variant from clinical prediction, the *intercept-only update* (Janssen et al., 2008), both fitted with balanced weights; the *top-q* rule, which flags in each cohort the share q of students with the highest scores, where q is either the at-risk rate of the training modules or, without labels, the prevalence that the EM procedure of Saerens et al. (2002) estimates in each cohort; and the *relative* model. All rules are evaluated on the same students. In a first scenario, $k = 100$ students drawn at random serve as the labelled sample (50 draws), and every rule is evaluated on the others. In a second, the first presentation of each module serves as a labelled pilot cohort, and every rule is evaluated on the later presentations. A top-q rule that uses the true at-risk rate gives a bound that no real deployment could reach.

We compute 95% confidence intervals with a cluster bootstrap over students (1,000 resamples; Efron & Tibshirani, 1993), which respects the dependence between the enrolments of a student in several modules. The same resamples are used for every configuration, so the intervals are paired. They describe the sampling of students, not of modules. At the module level, we therefore report how many modules improve, with a paired Wilcoxon test ($n = 7$), and we rerun both protocols with 10 seeds.

3.6 Interpretability and fairness

SHAP values (Lundberg & Lee, 2017) are computed with path-dependent TreeSHAP (Lundberg et al., 2020), so the attributions of a transferred model are relative to its training modules. We explain the transferred LOMO model and a *local* model trained on the module alone (two-fold cross-fitting, grouped by student) for 400 students per module, summing the attributions of one-hot columns back to their variable. The SHAP value of a forest is

Table 1. Methodological audit at the 25% cut-off (random forest). Each row adds one correction to the previous one. AUC of protocol (A), random split grouped by student, and (B), leave-one-module-out (LOMO), mean over held-out modules.

Evaluation setting	AUC (A)	AUC (B)	Gap
1) Reference protocol (all enrolments, pooled AUC in (A))	0.882	0.848	0.033
2) + early scores filtered on submission date	0.880	0.847	0.033
3) + students still enrolled at the cut-off only	0.815	0.787	0.028
4) + same aggregation in (A) and (B)	0.800	0.787	0.014
5) + context-relative features	0.816	0.808	0.008

estimated without bias from 100 randomly drawn trees, which reproduces the exact importance ranking (Spearman $\rho = 0.985$). The *agreement* between transferred and local models is the Spearman correlation between their global importances (mean $|\phi|$). The *drift decomposition* averages the signed attributions of the transferred model over the students S_m of module m and over feature families G,

$$\bar{\phi}_{m,G} = \frac{1}{|S_m|} \sum_{i \in S_m} \sum_{j \in G} \phi_{ij}. \tag{5}$$

Summed over families, these terms equal the mean difference between the predicted risk of m's students and the model's average prediction on its training modules.

We compare true-positive rate (TPR), false-positive rate (FPR), precision (PPV) and flag rate between women and men, between the most and least deprived IMD quintiles (Q1: 0–20%; Q5: 80–100%), and between students with and without a declared disability, over all modules and within each module. To examine composition effects, we regress the flag of students who are not at risk on gender, with and without module fixed effects (linear probability model). These are descriptive associations; they do not identify what causes a disparity.

All analyses were run in Python 3.12 with fixed seeds. One script regenerates every figure, table and number of the article and its supplementary material from the raw OULAD files, and automated tests check for temporal leakage and for students on both sides of either protocol. The code is available to reviewers on request and will be made public upon publication (link removed for double-blind review).

4. Results

4.1 Methodological audit

We started from a pipeline that keeps all enrolments and compares a pooled AUC in (A) with a per-module mean in (B). At 25% of the course, it gives an AUC of 0.882 in (A) and 0.848 in (B), a transfer gap of 0.033 (Table 1; gaps are computed before rounding). Filtering early scores by their submission date makes little difference. Restricting the population to students still enrolled at the cut-off has by far the largest effect. At 25% of the course, 19.5% of enrolments have already ended, including 62.5% of all Withdrawn outcomes and 2,678 unregistrations before the course had even started. Their last active day makes these students trivial to predict, and their share varies from 5.4% (GGG) to 28.0% (CCC), which distorts comparisons between modules. Using the same aggregation in both protocols then halves the gap again, from 0.028 to 0.014. Together, these choices make the apparent gap more than twice its corrected value. They also explain why our AUCs at 25% of the course, around 0.80, are lower than the values above 0.90 often reported on OULAD, which are usually obtained with features covering the whole course or with students who have already left.

4.2 How early can we predict?

Figure 1 plots performance against the length of the observation window. With relative features, the AUC increases from 0.749 at 10% of the course to 0.878 at 50% in (A), and from 0.739 to 0.871 in (B). The gap between the protocols stays between 0.007 and 0.010 at every cut-off (0.012–0.017 with absolute features). With absolute features,

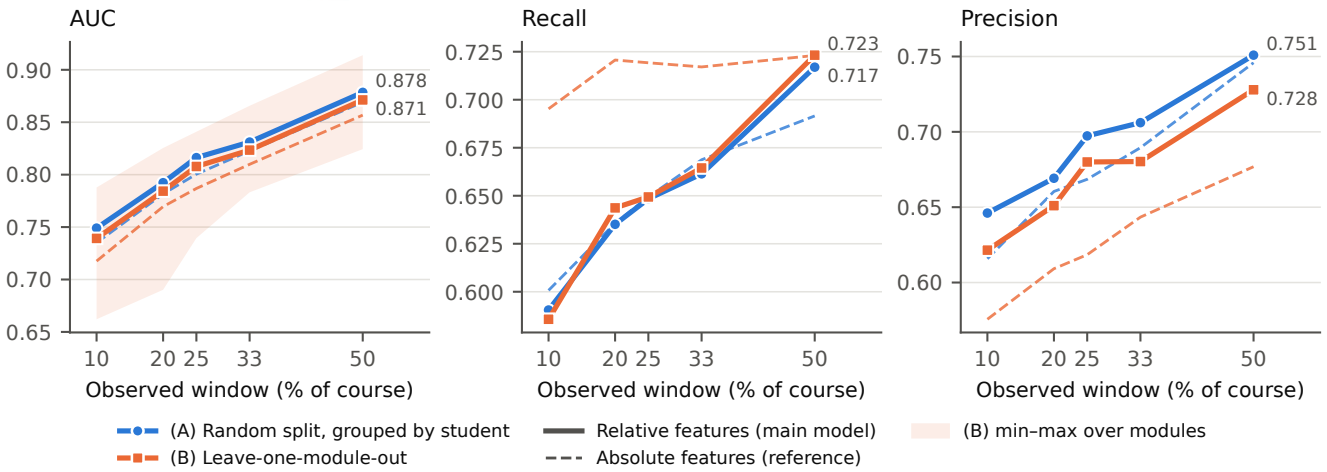


Figure 1. Performance as a function of the observed share of the presentation (students still enrolled at the cut-off). Solid lines show relative features and dashed lines absolute features. The band gives the range across held-out modules in (B) with relative features.

Table 2. Module characteristics at the 25% cut-off, students still enrolled at the cut-off. At-risk: Withdrawn or Fail. First deadline: first non-exam assessment, as a fraction of the presentation length. Exam weight: share of the exam in the total assessment weight. Early score: share of students with at least one assessment submitted before the cut-off. Values are averaged over the presentations of each module.

Module	n	At-risk (%)	Withdrawn among at-risk (%)	First deadline	Exam weight (%)	Early score (%)	Median clicks	Aged 35+ (%)
AAA	698	23.9	45.5	0.07	50	98.1	588	52.6
BBB	6,202	39.5	28.0	0.07	50	93.4	224	34.9
CCC	3,191	47.4	48.4	0.07	67	93.1	402	30.3
DDD	4,984	47.7	40.7	0.09	50	93.4	382	26.2
EEE	2,470	33.2	31.6	0.13	50	90.9	586	24.9
FFF	6,289	42.0	35.4	0.08	50	94.5	866	26.2
GGG	2,398	36.9	17.6	0.24	100	53.3	81	39.0

the transferred model flags too many students. At 25%, its recall is higher than in (A) (0.719 versus 0.649) but its precision is lower (0.618 versus 0.669). A study reporting recall alone, as is often done, would conclude that the transferred model is better than the local one.

4.3 Ranking transfers, alert volume does not

Table 3 and Figure 2 break down the three quantities per module at the 25% cut-off. The cost in ranking is small: 0.014 [0.010, 0.017] on average with absolute features, and less than 0.015 in 6 of the 7 modules. A naive reading of LOMO results would describe modules AAA and GGG as poorly transferable. They are in fact *difficult* modules. Their AUC is already low when the model has seen them (0.733 and 0.739), and it drops by only 0.002 and 0.014 when the model has not. BBB is the exception. Its cost of 0.043 persists within its presentations (0.033) and falls to 0.011 with relative features, which suggests that in BBB the ranking, too, depends on scale.

Alert volume behaves quite differently. In GGG, the transferred model flags 95% of students, compared with 44% for the model that has seen the module and an observed risk of 37%. Recall rises from 0.66 to 0.99, and precision falls from 0.55 to 0.38. BBB also drifts upwards (+0.19), while CCC and DDD drift slightly downwards. The mean absolute drift $|\Delta^{\text{same}}|$ is 0.135 [0.130, 0.141]. In practice, an institution that deployed this model in a course like GGG would end up contacting almost every student.

4.4 Calibration, not just the threshold

One might suspect that the drift simply reflects the choice of threshold. It does not. With absolute features, the mean drift stays between 0.084 and 0.138 for every threshold from 0.3 to 0.7, and the mean predicted risk, which involves no threshold at all, shifts by 0.075 on average and by +0.30 in GGG. Calibration worsens accordingly (Table S8 of the supplementary material). Moving to an unseen module changes the Brier score by 0.039 on average (+0.180 in GGG), calibration-in-the-large by 0.42 (−1.83 in GGG), and the calibration slope of GGG by −0.321. In

Table 3. Ranking versus threshold transfer per module (25% cut-off, students still enrolled). AUC and flag rate ($p \geq 0.5$) are computed on the students of each module; (A) uses out-of-fold predictions of the random split, (B) the model trained on the six other modules.

	Ranking (absolute features)			Flagged students, % (absolute)			Relative features, (B)	
Module	AUC (A)	AUC (B)	Cost	Observed risk	(A)	(B)	AUC	Flagged (%)
AAA	0.733	0.731	+0.002	24	35	45	0.792	37
BBB	0.796	0.753	+0.043	39	37	56	0.801	40
CCC	0.830	0.817	+0.012	47	44	41	0.811	46
DDD	0.837	0.827	+0.010	48	44	39	0.841	39
EEE	0.841	0.838	+0.003	33	26	31	0.837	29
FFF	0.828	0.817	+0.011	42	36	33	0.833	36
GGG	0.739	0.725	+0.014	37	44	95	0.739	30
Mean	0.800	0.787	+0.014	39	38	49	0.808	37

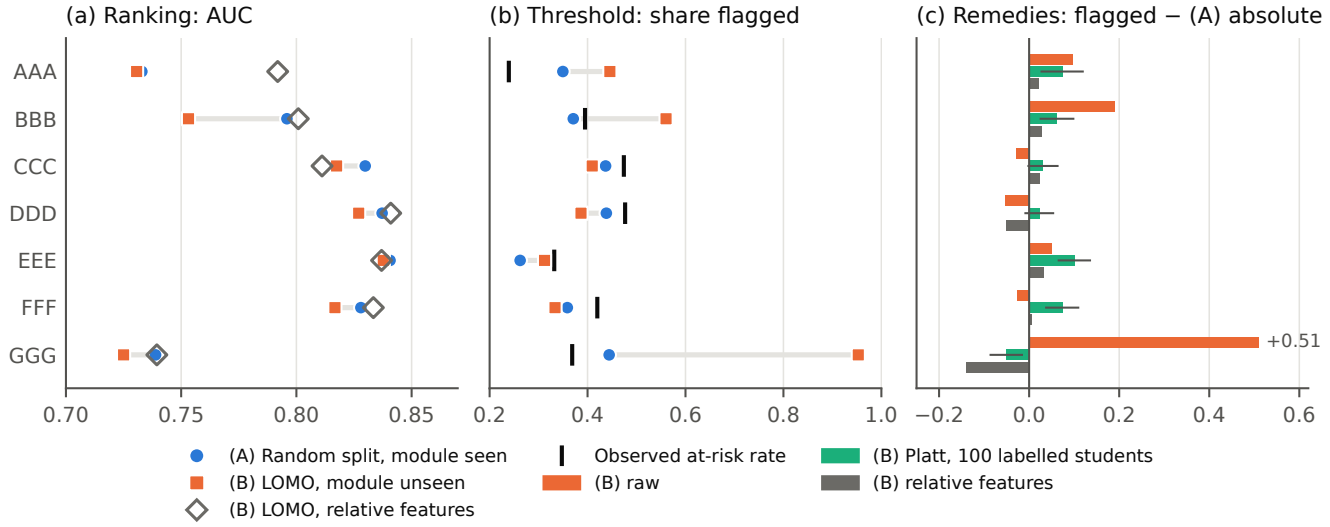


Figure 2. Ranking and alert volume per module at the 25% cut-off. (a) AUC when the module is seen (A) or unseen (B), and in (B) with relative features. (b) Share of students flagged in (A) and (B), compared with the observed at-risk rate. (c) Remedies applied to the same students, against the common reference (Δ^{ref}): raw transferred model, Platt recalibration on 100 labelled students (mean and standard deviation over 50 draws) and relative features.

the reliability diagrams (Figure 3), the three models are close to one another over all modules. In GGG, however, the transferred absolute model predicts risks well above those observed, whereas the relative model stays close to the local one. With relative features, the average changes shrink to 0.006, 0.08 and 0.077.

Nor are the balanced weights to blame. An unweighted forest calibrated by isotonic regression drifts just as much on unseen modules ($|\Delta^{\text{same}}| = 0.150$, with a mean predicted risk 0.139 away from the observed rate), and a logistic regression drifts as well, though less (0.061; Table 5).

4.5 What makes GGG different

Table 2 shows how GGG stands apart. Its first non-exam assessment comes at about a quarter of the presentation (7–13% elsewhere), its assignments are formative, and only half of its students have an early score at the cut-off. Its students also click much less: a median of 81 clicks before the cut-off, against 432 in the other modules. In the other modules, students with at most 81 clicks are at risk in 79% of cases, against 47% in GGG. Likewise, a student without an early score is at risk in 98% of cases elsewhere, but in only 51% of cases in GGG, where 47% of students have no score yet. The drift does not end with the first deadline: it reaches +0.60 at 33% and +0.51 at 50% of the course.

What the model has learned are absolute thresholds, such as “few clicks” or “no score yet” signalling risk, and the meaning of these thresholds depends on how a course is designed. Across presentations, the drift tends to be

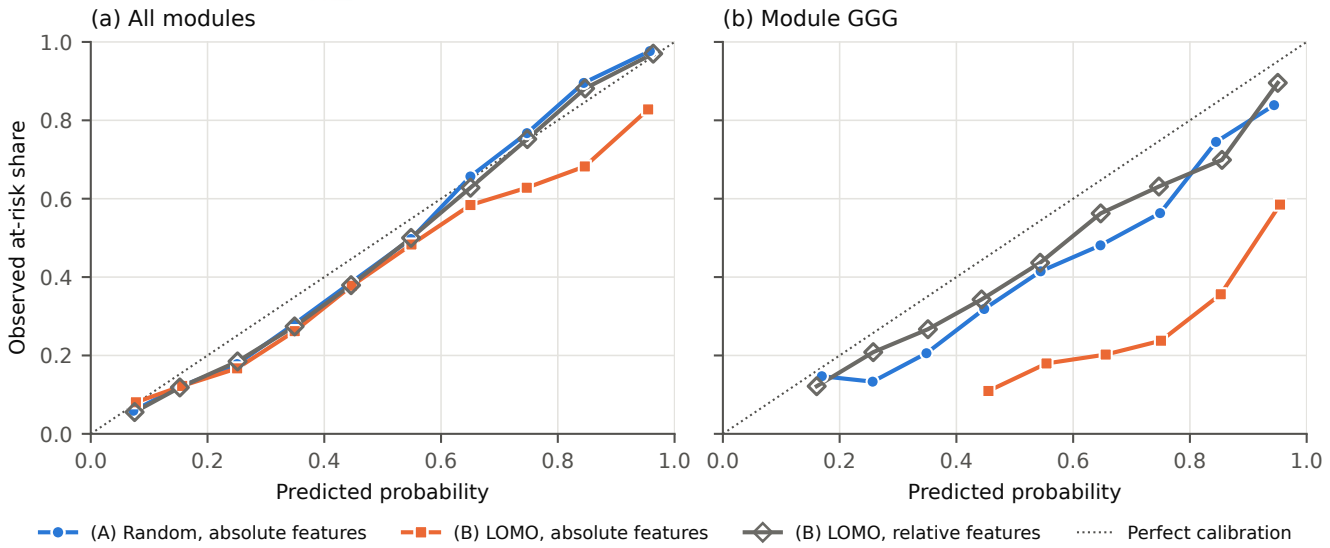


Figure 3. Reliability diagrams at the 25% cut-off: observed at-risk share for each class of predicted probability (classes with fewer than 30 students omitted). Because of the balanced class weights, the curves need not follow the diagonal even in (A); the effect of transfer is the gap between the (B) and (A) curves.

larger where at-risk students fail rather than withdraw (Spearman $\rho = -0.68$ with the share of withdrawals among at-risk students), where students are older and less qualified, and where the VLE is used less. With seven modules, these associations are only exploratory (Table S5 of the supplementary material).

4.6 Interpretability: the same features on different scales

If transfer failed because different features matter in different modules, the global importances of the transferred and local models would differ. Mostly, they do not. Local models rank the 18 variables similarly across modules (mean pairwise Spearman 0.81), with the mean early score at the top in every module. The agreement between transferred and local importances is moderate with absolute features (0.60) and higher with relative features (0.81), and it improves in 6 of the 7 modules (Figure 4a). Yet GGG, the module that drifts most, shows one of the *highest* agreements with absolute features (0.81): what goes wrong there is not the choice of features but the way they are read.

The drift decomposition of Equation (5) shows this directly (Figure 4b). For GGG's students, the transferred absolute model raises the predicted risk by +0.30 above its training average. Early-score features account for +0.19 and engagement features for +0.12, while demographics contribute −0.01. Across modules, this total follows the flag-rate drift ($\rho = 0.71$). With relative features, the shift largely disappears (−0.04 for GGG; Figure 4c). Because global importance ignores the sign of attributions, it cannot reveal a calibration problem, whereas signed attributions computed on the target population can.

4.7 Remedies compared on the same students

Table 4 compares the remedies on the same students. The label-free prevalence correction does not work: for GGG, the prevalence estimated by EM leads to flagging 100% of students, and on average the rule ends up further from the observed risk than the raw model ($|\text{flagged} - \text{risk}| = 0.224$ against 0.174). This fits our diagnosis. Under a pure change of prevalence, our balanced decision rule would need no correction; the drift comes from a change in how features relate to risk, which inflates the posteriors that prevalence estimation relies on.

With 100 labelled students, Platt recalibration and the intercept-only update both bring the alert volume close to the observed risk (0.057 and 0.058), but they barely change precision (0.624 and 0.627, against 0.619) and vary from one sample to the next (standard deviation of the share flagged: 0.039 and 0.040). Flagging a fixed share of each cohort reaches a distance of 0.068. The relative model, which uses no labels, comes as close to the observed risk as the two recalibrations (0.057, the same as the model that has seen the module, 0.059), with the highest precision (0.681) and F1 score (0.659). With a labelled pilot cohort instead of randomly drawn students, recalibration does worse (precision 0.615), because the presentations of a module also differ from one another, whereas the relative model keeps its advantage (0.679). Against the common reference, the relative model flags too few students in GGG

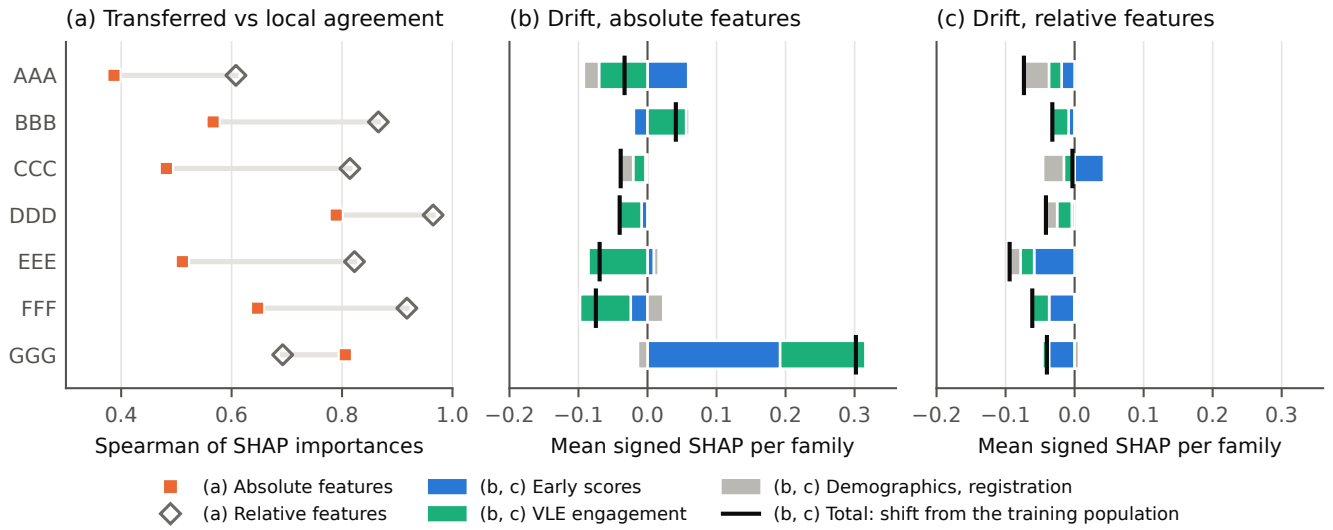


Figure 4. Feature importance and signed attributions at the 25% cut-off. (a) Spearman correlation between the SHAP importances of the transferred model and those of a model trained on the module itself. (b, c) Mean signed SHAP attribution of the transferred model for the students of each module, by feature family, that is, the shift of their predicted risk relative to the training population.

($\Delta^{\text{ref}} = -0.14$), but that reference flags too many there, and what matters for alert volume is the distance to the observed risk.

4.8 Choice of normalization and of learner

Percentile ranks are not the only way to normalize within the cohort (Table S9). On unseen modules, a winsorized z-score reaches an AUC of 0.801 and a robust standardization 0.800, compared with 0.808 for ranks and 0.787 for absolute values; over three cut-offs, the three normalizations differ by at most 0.008 in mean AUC. Normalizing within the cohort is what matters most; ranks are the best option or tied for best, and have no parameter. They do need a cohort of reasonable size. Computed within random sub-cohorts of the target module, they give an AUC on unseen modules of 0.778 [0.773, 0.781] with 10 students and 0.802 [0.800, 0.804] with 50, against 0.808 with the whole presentation (Table S2). From about 50 students onwards, ranks are as reliable as with the whole cohort.

The effect does not depend on the learning algorithm (Table 5). Ranks raise the AUC on unseen modules and reduce the drift for the logistic regression (AUC from 0.775 to 0.805, drift from 0.061 to 0.018) and for the calibrated unweighted forest (from 0.786 to 0.810, and from 0.150 to 0.034), as they do for gradient boosting and for a recurrent network on weekly activity sequences (Table S6). For the calibrated forest, the share flagged at 0.5 need not match the at-risk rate, since a calibrated model flags only the students more likely at risk than not; the relevant measure is then the distance between mean predicted risk and observed risk, which ranks reduce from 0.139 to 0.052. Average precision leads to the same conclusions (from 0.720 to 0.755 for the forest; Table S3).

4.9 Uncertainty, heterogeneity and robustness

For these students, the differences between relative and absolute features are precisely estimated: +0.021 [+0.015, +0.027] in AUC on unseen modules, −0.005 [−0.009, −0.001] in ranking cost and −0.115 [−0.120, −0.107] in mean absolute drift. At the module level, the picture is less uniform. The drift decreases in 6 of the 7 modules (Wilcoxon $p = 0.047$). The AUC improves in 5 ($p = 0.078$), so the gain in ranking is not established at the module level. The gain is large for AAA (+0.063) and BBB (+0.048), whereas CCC loses −0.008 and EEE −0.002 (means over 10 seeds; standard deviation across seeds at most 0.004 per module). CCC is the module whose absolute scales already transfer best (drift 0.027), and its loss persists within each presentation (−0.014): ranks discard part of the resolution of the absolute values, which was useful there, with nothing to correct.

The conclusions hold when the training sets have the same size (the transfer cost is then 0.014 with absolute and 0.008 with relative features), when the students shared with the held-out module are kept in training (mean change in AUC of at most 0.003), and when the threshold, the learner or the target definition is changed (supplementary material).

Table 4. Remedies compared on the same students (25% cut-off; means over modules). Random sample: Platt and intercept-only updates use $k = 100$ labelled students of the target module drawn at random (50 draws; standard deviation of the share flagged in parentheses), and all rules are evaluated on the remaining students. Pilot: the first presentation of each module is the labelled cohort and all rules are evaluated on later presentations. $|\Delta^{\text{ref}}|$: absolute flag-rate drift against the absolute-feature model with the module seen; |flagged − risk|: distance between the share flagged and the observed at-risk rate, which requires no reference model. Top-q flags the q highest scores of each cohort, q being the at-risk rate of the training modules, the rate estimated without labels by EM (Saerens et al., 2002), or the true rate (unreachable bound). Per-module values are given in the supplementary material.

	Target	Random sample ($k = 100$)				Pilot cohort			
Rule	labels	$\|\Delta^{\text{ref}}\|$	|flagged − risk|	Precision	F1	$\|\Delta^{\text{ref}}\|$	|flagged − risk|	Precision	F1
(A) absolute, module seen	—	0.000	0.059	0.669	0.653	0.000	0.077	0.666	0.650
Raw transferred	no	0.135	0.174	0.619	0.637	0.134	0.193	0.621	0.629
Platt recalibration	k	0.059 (0.039)	0.057	0.624	0.647	0.097	0.127	0.615	0.629
Intercept-only update	k	0.051 (0.040)	0.058	0.627	0.647	0.090	0.125	0.621	0.627
Top-q (training rate)	no	0.058	0.068	0.631	0.644	0.061	0.061	0.623	0.641
Top-q (EM estimate)	no	0.187	0.224	0.623	0.585	0.205	0.273	0.640	0.559
Relative features	no	0.043	0.057	0.681	0.659	0.067	0.066	0.679	0.655
Top-q (true rate, bound)	oracle	0.060	0.003	0.648	0.647	0.088	0.013	0.640	0.642

Table 5. Is the effect specific to the weighted forest? Protocol (B), 25% cut-off, means over modules, with absolute features and within-cohort ranks. Weighted forest and logistic regression use balanced class weights and threshold 0.5; the unweighted forest is calibrated by isotonic regression and uses threshold 0.5 on calibrated probabilities. AP: average precision. $|\Delta^{\text{same}}|$: absolute flag-rate drift against (A) with the same model and features. $|\bar{p}-\text{risk}|$: distance between mean predicted risk and observed at-risk rate. |ΔBrier|: absolute change of the Brier score from (A) to (B). Capacity precision: precision when flagging the 30% highest scores of each cohort.

Model	Features	AUC (B)	AP (B)	$\|\Delta^{\text{same}}\|$	|flagged − risk|	$\|\bar{p}-\text{risk}\|$	|ΔBrier|	Capacity precision
Weighted forest	absolute	0.787	0.720	0.135	0.174	0.134	0.039	0.708
	rank	0.808	0.755	0.021	0.057	0.069	0.006	0.731
Logistic regression	absolute	0.775	0.716	0.061	0.148	0.123	0.022	0.694
	rank	0.805	0.751	0.018	0.079	0.094	0.004	0.720
Unweighted calibrated forest	absolute	0.786	0.720	0.150	0.191	0.139	0.038	0.708
	rank	0.810	0.756	0.034	0.112	0.052	0.006	0.732

4.10 Fairness

With absolute features, transfer creates a gender gap that is absent when the modules are seen (Table 6 and Figure 5): the FPR gap between women and men goes from −0.007 [−0.019, +0.005] in (A) to +0.194 [+0.180, +0.208] in (B). Within modules, however, the transferred model's FPR is never significantly higher for women, and it is significantly lower for them in 5 of the 7 modules (Figure 5c). A regression with module fixed effects confirms that this is a composition effect, as the gender coefficient goes from +0.194 without fixed effects to −0.048 [−0.062, −0.033] with them. Women make up 88% of BBB and 81% of GGG, the two modules whose alert volume drifts upwards, so pooling the modules attributes their drift to gender (Simpson, 1951). With relative features, the pooled gap shrinks markedly, to +0.015 [+0.003, +0.026]; it is reduced, but not eliminated.

Gaps by IMD and disability behave differently. They appear under every protocol and reflect different base rates (51% at risk in Q1 against 33% in Q5). Unlike the gender gap, they are also present within courses: the FPR of Q1 is significantly higher in 5 of the 7 modules. Relative features reduce the pooled FPR gap by −0.039 [−0.064, −0.017] without closing it, which is consistent with the impossibility of equalizing both error rates and calibration when base rates differ (Chouldechova, 2017; Kleinberg et al., 2017). With absolute features, disadvantages accumulate at the intersections (FPR 0.496 for women in Q1 against 0.169 for men in Q5), whereas with relative features the remaining gaps follow deprivation rather than gender (Table S4).

5. Discussion

5.1 What transfers and what does not

Across modules, students are ordered by risk in much the same way: early scores, recent activity and submissions are associated with risk everywhere. What differs from one course to another is the scale on which these signals

Table 6. Between-group gaps in error rates at the 25% cut-off (threshold 0.5), with 95% confidence intervals from a student-level cluster bootstrap (1,000 resamples). TPR: recall; FPR: false-positive rate; PPV: precision; Flag: share of students flagged.

Contrast	Configuration	TPR gap	FPR gap	PPV gap	Flag-rate gap
Female − male	(A) absolute	−0.080 [−0.096, −0.061]	−0.007 [−0.019, +0.005]	−0.042 [−0.058, −0.025]	−0.050 [−0.062, −0.038]
	(B) absolute	+0.018 [+0.001, +0.035]	+0.194 [+0.180, +0.208]	−0.173 [−0.189, −0.155]	+0.111 [+0.099, +0.123]
	(B) relative	−0.065 [−0.082, −0.047]	+0.015 [+0.003, +0.026]	−0.060 [−0.077, −0.043]	−0.031 [−0.043, −0.020]
IMD Q1 − Q5	(A) absolute	+0.135 [+0.104, +0.165]	+0.148 [+0.127, +0.169]	+0.044 [+0.013, +0.077]	+0.224 [+0.205, +0.243]
	(B) absolute	+0.145 [+0.115, +0.175]	+0.156 [+0.132, +0.180]	+0.111 [+0.080, +0.142]	+0.219 [+0.200, +0.239]
	(B) relative	+0.125 [+0.095, +0.157]	+0.116 [+0.096, +0.137]	+0.062 [+0.032, +0.094]	+0.205 [+0.186, +0.226]
Disability yes − no	(A) absolute	+0.095 [+0.067, +0.121]	+0.102 [+0.076, +0.128]	+0.020 [−0.006, +0.047]	+0.146 [+0.124, +0.167]
	(B) absolute	+0.050 [+0.025, +0.074]	+0.111 [+0.083, +0.139]	+0.036 [+0.011, +0.064]	+0.121 [+0.099, +0.140]
	(B) relative	+0.078 [+0.051, +0.105]	+0.060 [+0.034, +0.085]	+0.045 [+0.020, +0.072]	+0.117 [+0.094, +0.138]

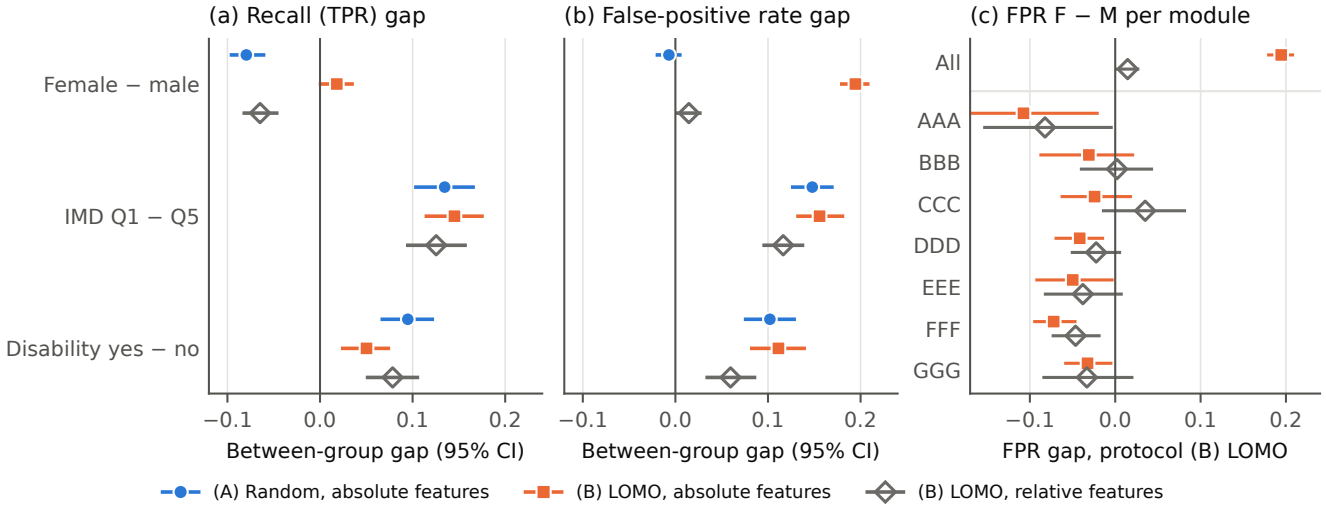


Figure 5. Gaps in error rates between groups (25% cut-off, 95% cluster-bootstrap intervals). (a) Recall gaps and (b) false-positive rate gaps. (c) Female–male FPR gap of the transferred model within each module and over all modules, with absolute and relative features; the pooled gap of the absolute model is a composition effect.

are expressed, for example how many clicks a typical week requires, when the first assessment is due, or whether early assignments are graded. A model that has learned absolute thresholds carries the scale of its training courses into the new course, and its probabilities degrade, much as discrimination travels better than calibration in clinical prediction (Janssen et al., 2008; Van Calster et al., 2019). The failure of prevalence correction supports this account: the at-risk rate does vary between modules (Table 2), but what changes most is what a given level of activity means.

The benefits of cohort normalization are established for the students of seven modules, but they are not uniform across modules: the drift is reduced in 6 of them, the gain in ranking is not established at the module level, and CCC is slightly worse off. That normalization reduces sensitivity to course design is an interpretation, supported by the mechanism, the SHAP decomposition and the consistency across learners.

5.2 Implications for practice

For an institution that wants to use a model in a course without labelled history, our results suggest four steps. First, normalize engagement and early-assessment features within the cohort at the date of prediction. This needs no outcome data but a cohort of about 50 students or more, so smaller groups should be pooled across sections of the same course. Second, compare the alert volume with the risk expected from the first weeks; a large gap is a reason to investigate, although without labels it cannot tell a miscalibrated model from a cohort that really is more at risk. Third, recalibrate once a first cohort has been labelled, keeping in mind that recalibration on about a hundred students is unstable. Fourth, decide how alerts will be used before deploying them. Gaps linked to deprivation and disability persist within courses, so alerts should lead to an offer of support reviewed by a person rather than to an automatic decision, and their distribution across groups should be monitored.

Reducing a disparity does not by itself make decisions fair. Cohort normalization also changes who is flagged within a course, and it hides any disengagement shared by the whole cohort. A practical design could combine a

cohort-relative model for individual students with an absolute indicator at the cohort level, whose decline would call for a pedagogical response rather than individual contacts.

5.3 Implications for evaluation practice

Early-prediction studies should restrict the population to students still enrolled at the prediction date and check for leakage with an invariance test. Course-level hold-outs should remove the held-out course's students from training, and protocols should be compared with the same aggregation. Alert rates and calibration belong next to discrimination, since recall alone rewards over-flagging. And fairness audits that pool contexts need within-context audits too, or a calibration problem can pass for discrimination and a real bias can be masked by composition.

5.4 Limitations and future work

The study relies on one institution and seven modules, so module-level tests have little power, presentations are not independent, and our bootstrap intervals reflect the sampling of students rather than of courses. Richer sequence models than those we tested might extract more temporal information, and SHAP values are estimated on a subsample of trees and students. Cohort normalization is transductive, needs cohorts of about 50 students and cannot detect a cohort that disengages as a whole. Our next step is to apply the same protocol to Moodle data from another, fully online university, where the held-out context will first be a degree programme and then a country, and to test the two-level design described above.

6. Conclusion

When the evaluation is strict (early in the course, restricted to students still enrolled, with no student on both sides of the split and with comparable aggregation), moving an at-risk model to an unseen course costs little in ranking but a great deal in alert volume and calibration. Calibration degrades mainly because engagement and early scores are read on absolute scales that depend on course design. The features that matter are the same in every course, and a change in prevalence is not the main cause. Normalizing these features within the cohort corrects the alert volume without labels, provided the cohorts are large enough, and it greatly reduces a gender disparity created by pooling courses. Early-warning systems meant for new courses should be evaluated on contexts they have not seen.

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